



Original Article

The genomic approach of antimicrobial resistance of *Salmonella* Typhimurium isolates from guinea pigs in Lima, Peru

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ABSTRACT

Salmonella Typhimurium is an important agent of foodborne diseases. In Peru, the emergence of multidrug-resistant isolates of *S. Typhimurium* from the food chain could be linked to guinea pig farming as a potential reservoir and their uncontrolled antibiotic treatment against salmonellosis. In this study, we performed the sequencing, genomic diversity, and characterization of resistance elements transmitted by isolates from farm and meat guinea pigs. The genomic diversity and antimicrobial resistance of *S. Typhimurium* isolates were performed using nucleotide similarity, cgMLST, serotyping, phylogenomic analyses, and characterization of resistance plasmids. We found at least four populations of isolates from farm guinea pigs and four populations from meat guinea pigs without finding isolated transmission between both resources. Genotypic resistance to antibiotics was observed in at least 50% of the isolates. Among the farm guinea pig isolates, ten were found to be resistant to nalidixic acid, and two isolates exhibited multidrug resistance to aminoglycosides, tetracycline-fluoroquinolone (carrying *strA-strB-tetA-tetB* genes and *gyrA* S83F mutation), or trimethoprim-sulfonamide (carrying *AaadA1-drfA15-sul1* genes). Additionally, two isolates from the meat source were resistant to fluoroquinolones (one of which had enrofloxacin resistance). The transmissible resistance plasmids with insertion sequences (IS) such as IncI-gamma-K1-ISE3-IS6, IncI1-I (alpha)-IS21-Tn10, and Col (pHAD28) were commonly found in isolates belonging to the HC100-9757 cluster from both guinea pigs and human hosts. Altogether, our work provides resistance determinants profiles and *Salmonella* sp. circulating lineages using WGS data that can promote better sanitary control and adequate antimicrobial prescription.

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1. Introduction

Guinea pig (*Cavia porcellus*) is a rodent species traditionally consumed in the Andean region, with a high nutritional value and economic impact on the countries of this region. In a worldwide context, Peru has the central production system of guinea pigs for human consumption. This system is predominantly familiar-commercial with an increasingly intensive scale [1]. Guinea pig

production has some limitations due to management and sanitary conditions deficiencies, inducing salmonellosis cases, a frequent disease associated mainly with the bacteria *S. Typhimurium* [1,2]. Furthermore, this infection affects guinea pigs' different systems (e.g., respiratory, digestive, nervous, and reproductive), which leads to a high morbidity and mortality rate [1] and turns this host into *Salmonella*'s transmission vector in the human food chain [3,4].

Salmonella Typhimurium is one the most important pathogens causing foodborne disease and salmonellosis in animal husbandry worldwide [5]. In Peru, the National Center of Epidemiology, Prevention, and Disease Control critically reported 1,754,586 diarrheal cases between 2018 and 2019 and 495 foodborne outbreaks between 2010 and 2021 [6,7]. *Salmonella* is the major contributor to the outbreak associated with 31 of them, with chicken and pork meat being the main food vehicle of transmission [7]. In the last year, the monitoring of guinea pigs with salmonellosis showed an incidence of 26% [1,8,9]. Few cases of Salmonellosis in humans are related to guinea pigs, being that seven cases were associated with consumed guinea pigs' meat between 2006 and 2009 in the USA [8].

The World Health Organization (WHO) lists *Salmonella* as a priority pathogen because of increasing multidrug resistance (MDR) cases, mainly to current treatment antibiotics (third-generation cephalosporin and fluoroquinolone). For animal husbandry, including guinea pigs, there is prophylactic consumption and indiscriminate use of antibiotics to treat salmonellosis [1], which is associated with the emergence of MDR strains and implies a risk to public health.

Mobile genetic elements (MGE) such as plasmids can rapidly disseminate antimicrobial resistance (AMR) traits through bacterial populations. Reconstruction and typing plasmids may provide insights into the spreading dynamics from livestock to humans [10]. Traditional epidemiological strategies such as microbiological, serological, and molecular analyses allowed the detection of scarce MDR lineages [11]. However, WGS-based specific strategies such as cgMLST, SNPs, pan-genome, phylogenomic, and GWAS would allow for the differentiation of closely related strains, detect outbreaks, predict antimicrobial susceptibility, and discover new AMR mechanisms [12,13].

Despite the emergence of foodborne outbreaks involving *Salmonella* and the increased antimicrobial resistance, there is no genomic tracking of the pathogen and antimicrobial resistance's risk transmission in guinea pigs' production. Currently, molecular and phenotypic strategies of *S. Typhimurium* isolates from guinea pigs farm found high resistance to first-line treatment antibiotics (chloramphenicol, tetracycline, sulfamethoxazole, nitrofurantoin, enrofloxacin, and quinolones) [3,14]. Meanwhile, farm surveillance worked with fewer representative source samples and traditional strategies (PCR, antibiogram) with a unique study at the genomic level. In this context, this study sequenced six novel genomes from guinea pig farms and added 20 genomes from food and farm guinea pigs to monitor and understand the genetic factors associated with antimicrobial resistance and transmission among *S. Typhimurium* isolates from guinea pigs.

2. Material and methods

2.1. Bacterial isolates

Six *S. Typhimurium* isolates were collected from guinea pigs with salmonellosis in two farms, as part of project "Caracterización del residuo fecal en animales de producción como amenaza potencial a la salud pública en Lima Metropolitana" in Lima, Peru. The isolates were processed at the Laboratory of Biology and Molecular Genetics, Faculty of Veterinary Medicine, Universidad Nacional

Mayor de San Marcos. Organ fragments were aseptically transferred to peptone water and incubated at 37 °C for 18 h. Afterward, the enriched cultures were transferred to Rappaport Vassiliadis Soya (RSV) broth and incubated at 42 °C for 24 h. Then, an aliquot was streaked in XLD (Xylose Lysine Deoxycholate) agar broth and incubated at 37 °C for 24 h. The *S. Typhimurium* colonies were confirmed by PCR reaction, according to Jamshidi [15]. The strains' details are described in Table 1.

2.2. Whole genome sequencing, assembly, annotation, and typing

Genomic DNA extractions were performed using PureLink™ Genomic DNA Mini Kit (Invitrogen, Waltham, MA, USA). DNA concentration was measured using Qubit dsDNA HS assay (Invitrogen). The genome sequence was performed on the HiSeq 2500 platform (Illumina, San Diego, CA, USA), and paired-end libraries (2 × 150 bp) were prepared using the Truseq3 library. The raw sequences were checked using FastQC v0.11.8 [16], and the trimming of low-quality reads was performed through Trimmomatic v0.39 [17].

For all isolates' sequences, a *de novo* assembly was performed using Unicycler v0.4.5 [18]. A hybrid assembly strategy was performed for the strain SE7, as follows. The SE7's scaffolds construction was performed using ContiguatorF [19] GenomeFinisher v1.4 (<http://gfinisher.sourceforge.net/index.php>). Strain SE7 was used as a Peruvian reference genome for subsequent analysis. The six genomes were submitted under the BioProject: PRJNA896238. For the subsequent analyses, the annotation was performed using PROKKA v1.11 [20]. Serovar identification was evaluated using SeqSero [21] and sequence type (ST) classification using MLST (<https://github.com/tseemann/mlst>).

2.3. Addition of 20 *Salmonella* Typhimurium genomes from food and farm guinea pigs

We added 10 *Salmonella* Typhimurium genome sequences isolated from two farm guinea pigs with salmonellosis from Lima, Peru, under BioProject PRJNA847586 obtained from NCBI. Other ten *Salmonella* Typhimurium were obtained as paired raw sequences isolated from guinea pigs' meat from Ecuador and Peru obtained from European Nucleotide Archive Database and included in the following analysis (Table S1). For the last 10 samples, the assembly, annotation, and typing were done following the method described in session 2.2.

2.4. Diversity analysis

All the orthologous genes across the 26 *S. Typhimurium* genomes were predicted using Roary v3.12 [21] with default parameters. The principal component analysis (PCA) was performed using the base R' `prcomp()` function to carry out the clustering of the strains by the accessory gene content. The phylogenetic tree was constructed based on the core genome using RAxML v8.2.12 [23] with maximum likelihood method, GTR + Gamma model, 1000 bootstrap replicates and visualized using the `ggtree` package in R (R Development Core Team, 2016). The core SNPs were obtained from the variant calling for all the isolates against the reference SE7 strain using `snippy` (<https://github.com/tseemann/snippy>). The nucleotide diversity between genomes was calculated using the pairwise SNP distances from the core-genome alignment using `SNP-dists` v0.6 (<https://github.com/tseemann/snp-dists>). We performed hierarchical clustering (HC) approaches of core genome MLST (cgMLST) using the Enterobase [24].

Table 1Epidemiological data, sequencing, genomic structure, and typing analyses from six *Salmonella* Typhimurium isolates from Peruvian Guinea Pigs.

Strain	Country	Source	Year	Accession number	Cov	Mb	# Contigs	# Proteins	# of Plasmids	Serovar typing	ST	H100
SE2	Peru	Farm	2016	JAPDVP000000000	55.26x	4811853	74	4490	2	Typhimurium	19	9757
SE3	Peru	Farm	2016	JAPDVO000000000	53.59x	4884534	72	4591	1	Typhimurium	19	9757
SE4	Peru	Farm	2016	JAPDVN000000000	55.15x	5014965	75	4741	2	Typhimurium	19	9757
SE5	Peru	Farm	2016	JAPDVM000000000	54.33x	4949980	76	4662	2	Typhimurium	19	9757
SE6	Peru	Farm	2016	JAPDVL000000000	55.03x	4841031	73	4541	2	4,[5],12:i:-	19	9460
SE7	Peru	Farm	2016	SUB12234903 (Biosample)	54.19x	4966101	2	4529	1	4,[5],12:i:-	19	9460

2.5. Prediction and characterization of mobilome in reference genomes

From the genomes we assembled, the plasmid contigs were separated from chromosomal contigs using MOB-suite [25]. The plasmids were annotated and classified by family and mobility using MOB-suite and oriTFinder [26]. Thus, we chose to circularize one reference sequence plasmid per family. Plasmid scaffolds' gaps were manually closed using consensus sequences generated by mapping reads against the reference plasmids (CP081672.1, CP016585.1, NZ_MK070495, NC_003277.2) using CLC Genomics Workbench v7.0 (Qiagen, USA). We evaluated nucleotide variant and phylogeny for each family plasmid following the steps and parameters described in method session 2.4.

Furthermore, prophages and genomic islands (GIs) were predicted from chromosomal reference SE7 using PHASTER (<https://phaster.ca/>) and islandViewer (<https://www.pathogenomics.sfu.ca/islandviewer/>), respectively. Insertion sequences (ISs) were predicted from reference chromosomes and plasmids using ISEScan [27]. Subsequently, the MGEs were mapped and compared at the nucleotide identity level against the other 25 strains using BRIG [28].

2.6. Identification of virulence and antimicrobial resistance genes

Antimicrobial resistance genes (ARGs) were predicted from chromosomes and plasmids of all strains using Abricate (<https://github.com/tseemann/abricate>) and CARD [29] database. The virulence genes prediction was performed using the VFDB database [30]. Minimum coverage and identity of 85% were used to identify those genes.

Chromosomal point mutations associated with AMR were predicted using PoinFinder [31]. The ARGs were classified according to resistance mechanisms and drug class using the CARD database and manual curation [29]. The ARGs and virulence genes were mapped on reference chromosomes and plasmids using BRIG to exhibit the architecture and repertoire genes of each MGEs.

2.7. Effect of new single nucleotide variations in antibiotic resistance proteins

ARGs sequences were extracted using Roary [22], followed by sequence alignment with MAFFT v7.307 [32] to identify single nucleotide variations (SNVs) in chromosomal ARGs. Then, SNP-sites [33] was used for the variants calling from gene alignments. Structure modeling of ARGs was performed and validated using the Swiss Model homology server and Ramachandran plot. The 3D structure was used to predict the stability of protein with an altered amino acid using the Dynamute2 tool [34].

3. Results

3.1. Diversity and phylogeny of *S. Typhimurium* isolate obtained from guinea pigs

The six *S. Typhimurium* isolates from Peruvian guinea pigs had a genome size between 4.81 and 5.01 Mb, with an average of 4607 proteins and one or two plasmids. The assembly quality has a mean depth coverage of >53.59x. MLST typing classified all strains belonging to ST19 (Table 1). For the following comparative analyses, we added 20 genome isolates from guinea pigs' farms and meat (Table S1).

We worked at the hierarchical clustering levels 100 (HC100) and 50 (HC50) to observe differences between phylogroups. The HC100 level comprised three clusters: HC100_9757 ($n = 14$) contains farm and meat samples from Peru and Ecuador. HC100_41507 contains meat samples from Ecuador ($n = 6$). HC100_9460 contains Peruvian samples from the farm ($n = 4$) and two samples from Ecuador and meat. The HC50 level comprised seven distinct sub-clusters or populations, and Peruvian isolates belong to the largest sub-clusters H50_67422, H50_9757, and HC50_109907 following HC50_61863 and Equatorial isolates belong to the largest cluster/sub-cluster HC100_41507/HC50_41507 following sub-cluster HC50_21802 and HC50_6907 (Fig. 1). Previously, the Peruvian HC100_9757 and HC100_9460 clusters were identified within the phylogenetic alpha clade to *S. Typhimurium* [35].

Genomic serotyping classified 19 isolates as serovar Typhimurium belonging to clusters HC100_9757 and HC100_41507. The other six isolates were classified as 4,[5],12:i:-, a monophasic serovar Typhimurium variant belonging to the HC100_9460 cluster (Fig. 1). The un-rooted phylogenetic tree is based on a core genome containing five sub-clades; the three most differentiated sub-clades correspond to three different HC100 clusters (Fig. 1). Some sub-clades harbor isolates with the same country or source origin, but they do not allow total discrimination. Four isolates from meat sources are most closely related to the two clusters from farm sources, FDA00011664 and FDA00000688 isolates belong to HC100_9757; FDA00011660 and FDA00000345 strains belong to HC100_9460 cluster (Fig. 1); however, the strains belong to different sub-clusters or populations.

The pan-genome analysis examined gene content diversity, and the core genome was used as input data to analyze the phylogeny. The pangenome has 5357 orthologous genes, 4205 core genes, and 1152 accessory genes. Core and accessory genomes constitute 78.49% and 21.50% of the repertoire genome, respectively. The accessory heatmap and PCA analysis with 49.52% of the gene repertoire distinguished the gene content in at least four groups, corresponding to HC50_9757, HC50_67422, HC100_41507, HC100_9460 (Fig. S1). The isolates from cluster HC100_41507 had fewer accessory genes from food samples compared to the

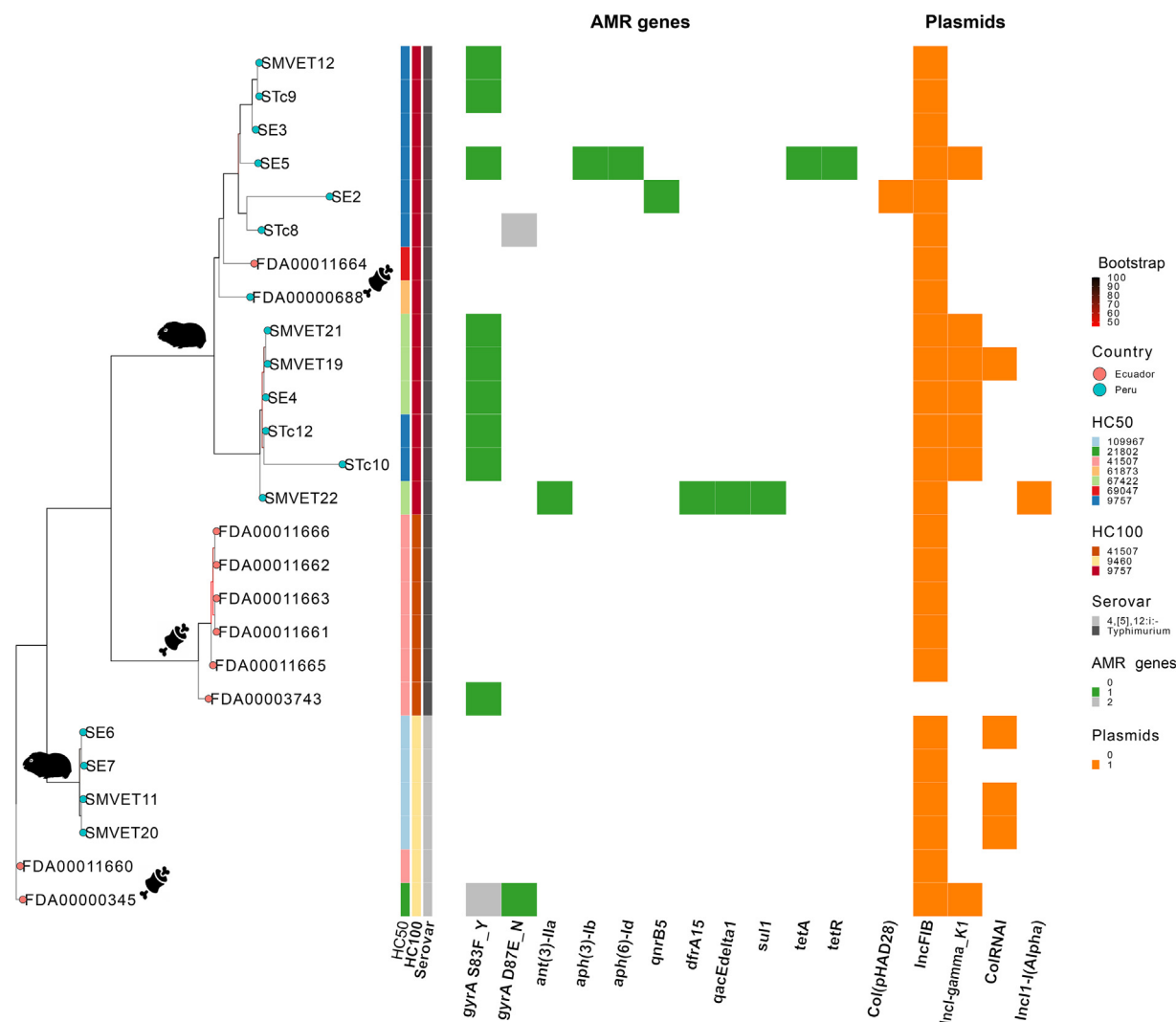


Fig. 1. The phylogenetic tree was generated using the ML method, 1000 replicates, and the GTR + GAMMA model to estimate the evolutionary divergence between 26 *S. Typhimurium* isolates from the farm guinea pigs and guinea pigs' meat. The heatmap shows the presence and absence of antimicrobial resistance chromosomal mutation and genes and plasmids for each isolate. Country, serovar, sources, and HC cluster were shown in the heatmap. HC100_9460 cluster strains were classified as serovar 4,5,12:i:-. The HC100_9757 (FDA00011664 and FDA00000688) and HC100_9460 (FDA00011660 and FDA00000345) cluster strains from meat sources were the most closely related to isolates from farm sources. FDA00011664, FDA00000688, FDA00011660, and FDA00000345 isolates from meat sources belong to the major HC100_9757 and HC100_9460 clusters but distinct sub-cluster or populations from farm sources.

H50_67422 sub-cluster and STc12 and STc10 strains (HC100_9757 cluster). The closest and largest sub-cluster and cluster distances were HC50_674222 (2–14 SNPs differences) and HC50_9757 (3–92 SNPs) from the HC100_9757 cluster; HC50_109907 (4–14 SNPs) from HC100_9460 cluster and HC100_41507 (1–53 SNPs) (Fig. S2).

3.2. Nucleotide similarity and mobile genetic elements shared between *Salmonella Typhimurium* strain SE7 and the others

Mobile genetic elements in SE7 chromosome were characterized by virulence, resistance, and phage content and compared with the chromosomes from the other 25 strains (Fig. 2). The comparison allowed us to identify shared and unique MGEs to distinct resistance and virulence regions and their possible transmission mechanism.

All strains share 18 genomic islands (GIs) and 156 virulence genes, which evidence the conservation of these MGEs and virulence in *S. Typhimurium* isolates. The 13 *Salmonella* pathogenicity islands (SPIs) and three phages carrying virulence genes were

characterized (Fig. 2). Among these, there are islands containing operons involved with membrane-associated type III secretion system (T3SS) (SPI1), intracellular survival (SPI2), intra-phagosomal survival and adhesion (SPI3), membrane ruffling and invasion (SPI5), T4SS secretion systems (SPI6), virulence and adhesion (SPI9), and cytoskeleton rearrangement (SPI12). Also, three SPIs contained the operons fimbriae/pili *fimAICDHFZYW*, *csgABCDEF*, and *lpfABCDE* (Table S3, Table S4).

We found seven prophages in all strains, including intact (Gifsy2, Gifsy1), questionable (Fels-1, Cronob_ESSI/ΦSE7-6), and incomplete (Cronob_vB/ΦSE7-1, Escher_500465/ΦSE7-4, Burkho_phiE255/ΦSE7-7) prophages (Fig. 2). The description of the prophages can be found in Table S4. Nucleotide similarity shows that the virulence prophage Cronob_ESSI is absent in the isolates from the HC100_9757 cluster. In addition, the adherence gene *shdA* was only present in strains SE3, STc10, and FDA00003743 (Table S3). Also, we identified twenty-seven ISs belonging to families IS3, ISNCY, IS200/IS605, IS21, IS66, IS256, IS630, IS5, and newIS contained mostly in prophages and GEIs (Fig. 2).

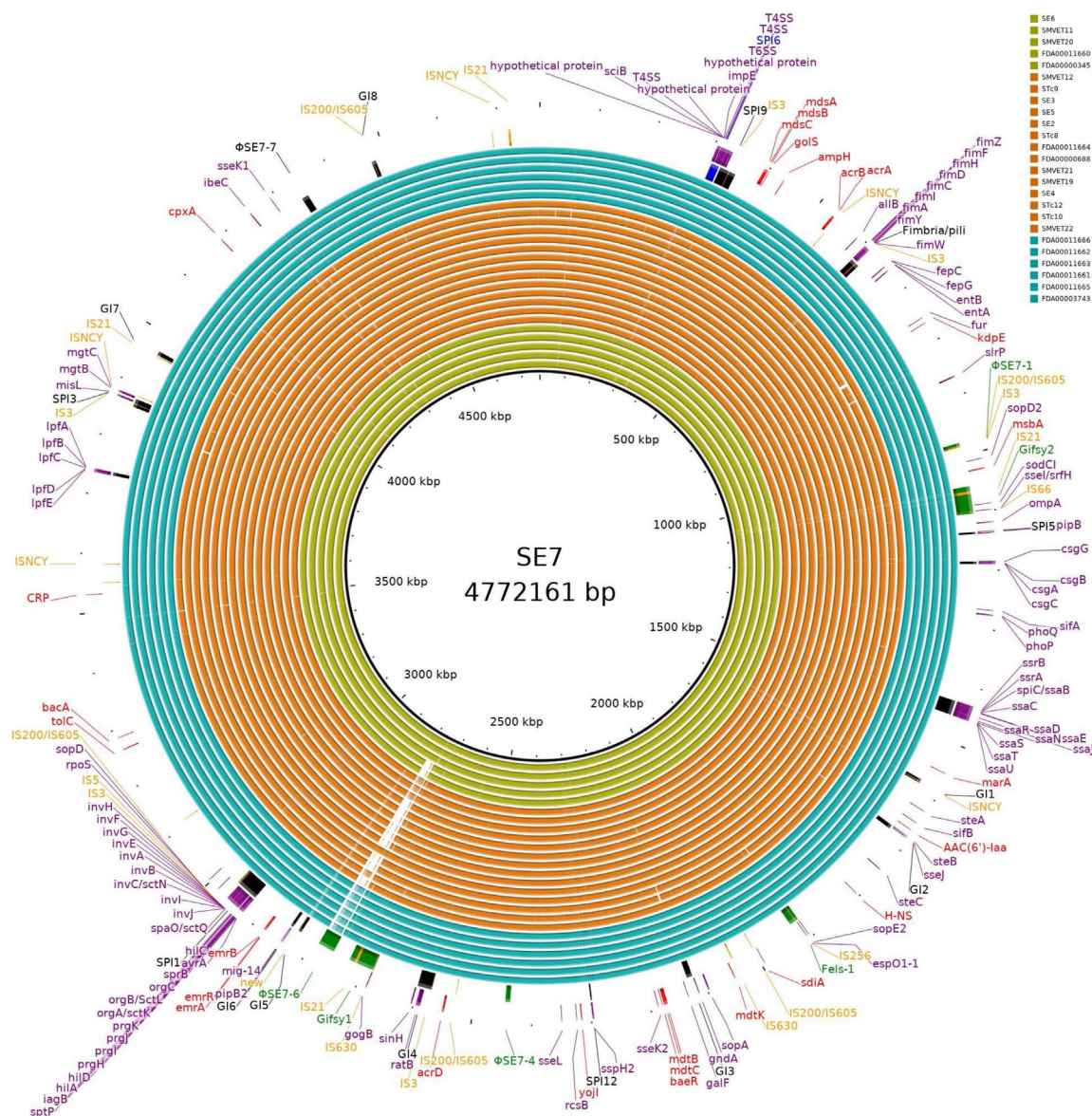


Fig. 2. Circular map and genome comparison of 26 *S. Typhimurium* isolates using SE7 as reference chromosome. The outer arch represents the location of 18 genomic islands (black), 7 prophages (green), 27 insertion elements (orange), 27 ARGs (red), and 156 virulence genes (purple). The first ring shows the coordinates and total genome size of the SE7 reference sequence, and the following rings 2 to 25 represent the comparison identity by intensity color against the other genomes. The colors of these rings represent the clusters HC100_9460 (green), HC100_9757 (orange), and HC100_41507 (blue). Exceptionally, nucleotide similarity shows the prophage region Cronob_ESSI not shared by all strains.

We characterized all the virulence and resistance plasmids and compared them to one plasmid reference for each plasmid family. Thus, we mapped the genes of interest on the reference plasmids p1-IncFIB-SE7, p2-ColRNAI-SE6, p2-IncI-gamma-K1-SE5, p2-IncI1-(alpha)-SMVET22, and p2-Col (pHAD28)-SE2. All the plasmids were mobilizable or conjugative; the three last plasmid families were present in isolates SE5, SMVET22, and SE2 belonging to the HC100-9757 cluster and carrying ARGs. All strains except FDA00003743 had a virulence plasmid belonging to the IncFIB family (Fig. S3), a prevalent plasmid found in the serovar Typhimurium. This plasmid shows a conservative region harboring *pefABCD*, *spvBCD* operon, *mig-5*, and *rck* genes associated with virulence without carrying resistance genes. This plasmid also harbored the type IV pilus genes involved in their conjugative transference. In addition, 43 SNPs were found between the 25 IncFIB plasmids, which include 21 missense variants, 11 synonymous variants, two-stop gains, and

non-coding. There are between 1 and 18 SNPs differences between HC100 intra-cluster plasmids (Fig. S4). Phylogenetic tree and nucleotide diversity based on 25 IncFIB plasmids show similar tree topology and divergence based on the core genome (Fig. S5).

3.3. Genotype profile of AMR acquired genes in plasmids and point chromosomal mutation

The antimicrobial resistance genotypic profile of the 26 isolates shows the presence and drug classification of 36 different ARGs, of which 28 were chromosomal, and eight were acquired genes (Table S2). Chromosomal ARGs were associated with resistance to aminoglycoside and/or aminocoumarin (AAC(6')-Iaa, *acrD*, *kdpE*, *mdtB*, *mdtC*, *baeR*, *cpxA*), fluoroquinolone (*emrABR*, *mdtK*, *gyrA*), and nitroimidazole (*msbA*). Others were beta-lactamase (*ompH*, *ompA*) and multiclass (*acrB*, *golS*, *marA*, *mdsA*, *mdsB*, *mdsC*, *CRP*, *acrA*, *acrE*,

acrF, *H-NS*, *sdhA*, *tolC*, *yojI*). The gene *ampH* associated with beta-lactam resistance was found in 50% of strains (Table S3).

Mobile genetic elements are associated with the antimicrobial resistance transmissible in bacteria. Eight acquired plasmid carrying ARGs were observed in only three strains with the potential for resistance transmission. Four isolates contained a small mobilizable ColRNAI plasmid carrying mobilization genes (*mbeA* and *mbeC*) and fewer ARGs (Fig. 3A). Seven strains carry the conjugative IncI-gamma-K1 plasmid of 97,882 bp of size and content type IV pilus genes (Fig. 3B). Only the plasmid p2-IncI-gamma-K1-SE5 included a region with ARGs encoding resistance against aminoglycosides (*strA*, *strB*) and tetracyclines (*tetA* and *tetR*) (Fig. 3B). It was also identified that the IS3 and IS6 families flanking the ARGs. This plasmid presented 2627 SNPs of 97,882 bp with a mutation rate of 2.68% (Fig. S6), including 1003 missense_variant, 1240 synonymous_variant, and 4 start_lost and

splice_region_variant&stop_retained_variant. The phylogenetic tree among the seven IncI-gamma-K1 plasmids shows major phylogenetic distance to p2-IncI-gamma-K1-SE5 (1535 SNPs differences) and p2-IncI-gamma-K1-FDA00000345 (1111 SNPs differences) (Fig. S7).

A unique conjugative p2-IncI-I (alpha)-SMVET22 plasmid of 101,676 bp was found in the SMVET22 strain carrying type IV pilus genes. This plasmid harbors ARGs encoding resistance against aminoglycosides (*ANT(3'')-Iia*), trimethoprim, and sulfonamide (*dfrA15* and *sul1*) (Fig. 3C) and is proximate to IS21 and Tn10 sites. A unique small mobilizable p2-Col (pHAD28)-SE2 plasmid of 2699 bp is found in the SE2 strain carrying the *qnrB4* gene that encodes fluoroquinolone resistance (Fig. 3D). Likewise, we identified different plasmids profile per cluster; in strains belonging to the HC100_9757 cluster, IncI-gamma-K1, IncI-I (alpha), and Col (pHAD28) plasmids were found; in the HC100_9460 cluster strains

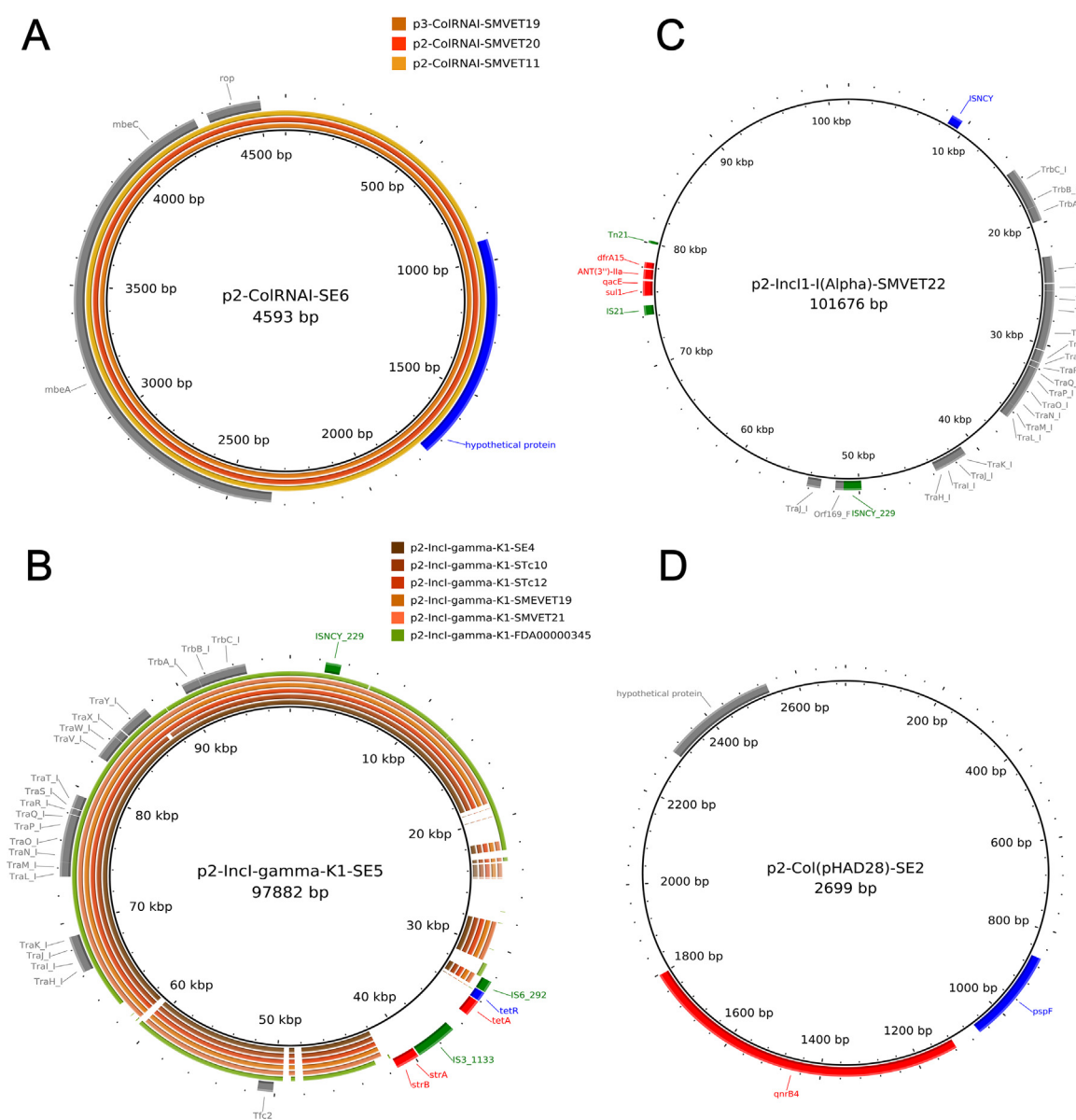


Fig. 3. A. Genome comparison of ColRNAI family plasmid. Three ColRNAI plasmid sequences were aligned to the reference p2-ColRNAI-SE6 plasmid. B. Genome comparison and repertoire genes of IncI-gamma-K1 family plasmid. Six conjugative IncI-gamma-K1 plasmid sequences were aligned to the reference p2-IncI-gamma-K1-SE5 plasmid. C. Genome structure of a unique IncI-I (alpha) plasmid of SMVET22 strain. D. Genome structure of a unique Col (pHAD28) plasmid of SE2 strain. The intensity of ring colors represents nucleotide similarity. The outermost arch indicates the location of T4SS genes (yellow), ARGs (red), insertion sequences (green), and virulence genes (purple).

presented commonly colRNAi, and in the case of HC100_41507 cluster strains did not present resistance plasmids.

Condition environmental and/or mutation could influence changes in the expression of chromosomal ARGs. As a result, we detected four different chromosomal mutations in the *gyrA* gene associated with fluoroquinolone resistance in 11 isolates. Nine isolates belong to the cluster HC100_9757 ($n = 8$) and HC100_41507 ($n = 1$) and present the most common mutation at codon 83 in the *gyrA* gene (S83F). Also, we detect the FDA00000345 (HC100_9460) strain carrying two mutations in *gyrA* p.S83Y and p.D87E associated with resistance ciprofloxacin/enrofloxacin; and STc8 (HC100_9757) isolate harbor mutation in *gyrA* p.D87N (Fig. 1).

Ten novel nonsynonymous SNVs were predicted in six AMR chromosomal genes (Table S5). In one isolate, two novel mutations were found in *acrE2* (G299D) and *acrF* (G929D); both genes were related to cephalosporin, carbapenem, and fluoroquinolone resistance. Also, novel mutations were found in *mtdB*, *mdtC*, *mdtK*, *aac(6'')-Iaa* contented by HC100 cluster lineages (Table S5).

3.4. Prediction of the effects of novel SNVs in antimicrobial resistance core genes

We evaluated the effect of the ten novel SNVs found on chromosomal ARGs based on their location within protein domains and alteration of protein thermodynamic stability. We focus on the nonsynonymous SNVs that could significantly alter the folding free energy, decrease the stability and affect the function of the native protein. First, we confirmed that five SNVs were found in domains of AcrE, AcrF, MdtC, and MdtK proteins (Table S5). However, we considered only two SNVs in the *acrE* and *acrF* genes due to their presence within the protein domain and causing destabilizing effects.

4. Discussion

The emergence of multidrug-resistant *S. Typhimurium* is a threat to treatments of salmonellosis in guinea pigs' husbandry, and these hosts are potential zoonotic reservoirs. Our study performed genomic surveillance to monitor lineages and antimicrobial resistance determinants circulating in isolates from guinea pig production.

4.1. Three major Peruvian populations associated with isolates from farm sources

S. Typhimurium isolates were classified as sequence type 19, the most frequent ST belonging to the serovar Typhimurium in clinical cases [12] and farm guinea pigs [35] in Peru. We reported range differences of 174–740 SNPs between the Peruvian HC100 cluster. A minor range of SNPs differences was reported in ten isolates of guinea pigs farm isolates included in our study [35]. Pan-genome analysis of large *S. Typhimurium* datasets from clinical and diverse host isolates showed a higher number of orthologous genes ($n = 7,462, 8,849, 6988$) and fewer core genes ($n = 3,181, 3,862, 4186$) [12,36,37] compared to gene variability of 26 guinea pigs' isolates.

Serotyping classified six isolates from meat and farm origin as the monophasic variant Typhimurium. It is associated with the absence or deletion of the flagellin genes (*fljA* and *fljB*) [38]. Isolates of *Salmonella* Typhimurium variant 4,[5],12:i:- were a successful global pathogen associated with foodborne outbreaks and linked to MGEs accumulation without impairment of bacterial fitness [39]. Isolates of this variant belong to the HC100_9460 cluster, with fewer ARGs and MGEs diversity with the presence of Cronob_ESS12 virulent phage.

This study reported three main clade-cluster/sub-cluster (H50_67422, H50_9757, and HC100_9460/HC50_109967) with

nucleotides and gene differentiation circulating in isolates from farm guinea pigs in Peru. These populations belong to the most genetically diverse α clade with isolates related to epidemics and transmission of AMR [37]. Previously, molecular and phenotypic studies identified resistant *S. Typhimurium* isolates belonging to at least a unique or seven populations circulating in farms guinea pigs in Lima [3,14]. Current population structure and HierCC-cgMLST analyses of primarily clinical samples identified ten lineages [12] and two HC100 clusters from farm guinea pigs samples in Peru [35].

The effective vector transmission of *Salmonella* to humans is recurrently reported between foods from intensive farming (poultry, pigs, cattle) and humans' outbreak [5]. The foodborne illness outbreaks were typically associated with dominant multi-drug clones like DT104, ST34, and ST313 linked to MGEs carrying ARGs [40,41]. Although four samples from the food source belong to the largest clades (HC100_9757 and HC100_9460), they belong to other sub-cluster or populations, so we still cannot define the transmission route for breeding and meat consumption of guinea pigs, which requires large and representative data.

4.2. Resistance and virulence in *Salmonella Typhimurium* isolated from guinea pigs

Thirteen SPIs are responsible for the invasion, survival, and establishment of systemic infection were predicted in all isolates and annotated according to Kombade and Kaur [42]. The 155 virulence genes were present in all isolates except the gene *sdhA*, present in only three strains. Previously, SPIs were not exclusively found in isolates causing salmonellosis in guinea pigs [35]. As a result, we did not find a differential content of genomic and pathogenic islands between isolates of guinea pigs with Salmonellosis.

Bacteriophages constitute a significant part of the accessory genome and allow distinct pathovariants and lineages within serovar Typhimurium. Of seven prophages reported Gifsy 1, Gifsy2, and Fels-1 are virulent and were involved in the intracellular survival of *S. Typhimurium* [35,37,43]. *Cronobacter sakazakii* phage ESS1-2 (Φ SE7-6) was detected mostly in HC100_9460 and HC100_41507 cluster strains, considered a virulent phage belonging to the family Myoviridae [44]. Unlike the high diversity *Salmonella* based on phages, this reduced number would be due to comparison based on a reference strain.

MGEs carrying ISs could mobilize and act in bacterial adaptation as resistance, virulence, and catabolism. Eight SPIs and virulent prophages were detected carrying the IS21, IS66, IS630, IS256, and frequently IS3. IS256 family members are known to modulate virulence in Gram-positive bacteria [45]. Insertion sequences IS21 and IS630 integrated into Gifsy1 could degrade or resuscitate prophages [43].

Likewise, we identified a large number ($n = 28$) of intrinsic ARGs in which environmental conditions would modulate their expression. Intriguingly, most ARGs encoded efflux pumps are associated with resistance to multiclass antibiotics following fluoroquinolone and beta-lactam resistance. The gene *ampH* was found in 50% of strains without domain mutations and was previously reported in Peruvian clinical strains [12]. *AmpH* is a penicillin-binding protein (PBP). Mutations in a PBP gene decrease protein affinity and low susceptibility for β -lactams in *S. Typhimurium* [46]. The conservative number of ARGs to NTS and Typhimurium strains so far shows their intrinsic resistance [47].

4.3. The presence of family plasmids and their SNPs and genes content could be a sign of resistance, origin source, and spread

The IncFIB virulence plasmid carrying the *spvBCD* and *pefABCD* operons was shared by 96% of isolates from guinea pigs. This

conserved plasmid is present mainly in serovar Typhimurium, in guinea pigs, and in clinical samples in Peru [12,35]. Furthermore, we emphasize 43 SNPs between the 24 IncFIB plasmids, including 21 missense variants that could act over virulence genes. In this way, IncFIB plasmid sequences allowed phylogenetic and nucleotide diversity distinction.

Also, we reported a high nucleotide variability in the IncI-gamma-K1 plasmid with an SNPs rate of 2.68%; these variants were presented in 45 CDSs but not in ARGs. Thus, increased missense variants could alter the function and offer a fitness cost. Few studies have performed in-depth analyses; a large dataset of the plasmids of *Chlamydia trachomatis* strains reveals a high SNPs rate of 2.97% in CDSs. However, these events could not be supported functionally or associated with dominant clones; additional evaluation should be considered [48]. The p2-IncI-gamma-K1-SE5 resistance plasmid is phylogenetically distant and could be submitted to evolutionary pressure distinctively. Unpublished data of phylogenetic analysis of IncI1-I (alpha), IncI-gamma-K1, and col (pHAD28) plasmids showed a closer relationship among resistant plasmids from guinea pigs and human hosts.

4.4. Reported a reduced resistance to first-line antibiotics mediated by plasmids and moderate nalidixic acid resistance by chromosomal mutations in *S. Typhimurium* isolates from guinea pigs in Peru

Between 2003 and 2007, Latin American countries reported 25 *Salmonella* outbreaks linked to animal-based foods and related to resistance to several antibiotics [4]. The consequent indiscriminate use of antibiotics as prophylactic and therapeutics in animal production, including guinea pigs, are a potential risk to antimicrobial resistance transmission and a threat to public health [49]. *Salmonella* isolates from farm guinea pigs show phenotypically high resistance to nitrofurantoin, erythromycin or sulfamethoxazole, enrofloxacin, and moderate resistance to amoxicillin, ciprofloxacin, and chloramphenicol [3,14]. We detected three transmissible family plasmids IncI1-I (alpha), IncI-gamma-K1, col (pHAD28). In Peru, these plasmid families were previously reported in *S. Typhimurium* isolates from guinea pigs and human hosts [12,35].

Genotypically, we reported 11 (42.3%) *S. Typhimurium* associated with nalidixic acid resistance, one isolate resistant to ciprofloxacin/enrofloxacin, and two MDR strains to aminoglycoside, tetracycline, and nalidixic acid or trimethoprim-sulfamethoxazole and aminoglycoside. Thus, compared to livestock and clinical samples, we present a reduced resistance genotypic profile to the first-line antibiotic of *S. Typhimurium* isolates from guinea pig hosts.

We reported a reduced resistance to the first-line antibiotic and high resistance to ciprofloxacin compared to small datasets of *Salmonella* isolates from farm guinea pigs and moderate MDR (12%) in large datasets from clinical isolates in Peru [12]. In Latin America, the USA, and England, NTS and *S. Typhimurium* strains of large datasets from foods and clinical cases showed a significant number (24.3%–43.0%) of MDR strains, including resistance to nalidixic acid, tetracycline, ampicillin, trimethoprim-sulfamethoxazole, streptomycin, chloramphenicol, and ciprofloxacin (in some cases) [5,50,51].

In the present work, we related three resistance transmissible elements that potentially would increase ARGs spread. We found the p2-IncI1-I (alpha)-SMVET22 linked to Tn21 carrying ARGs to aminoglycoside (*aadA1*) and trimethoprim-sulfamethoxazole (*dfrA*, *sul1*), already reported by Carhuarica [35]. Tn21 and IncI1 families are strongly mobilizing ARGs [52], such as spectrum-extended β -lactamases (ESBLs) linked to IncI1-I (alpha) carried in *S. Typhimurium* from clinical samples [9]. Interestingly, the p2-IncI-gamma-

K1-SE5 conjugative plasmid linked to ISE3 carries ARGs against aminoglycoside (*strA*, *strB*) and tetracycline (*tetA* and *tetR*). The IncI-gamma-K1 plasmid harbored a wide range of ARGs, and ESBL genes [53] and was associated with chloramphenicol and trimethoprim [9], but did not carry ARGs in isolates from guinea pig [35].

Fluoroquinolone resistance was associated with mutations in *gyrA* S83Y, D87E, D87N, and *qnrB4* gene carried in the Col (pHAD28)-SE2 plasmid. Previously, like genotypic profile was associated with nalidixic acid and ciprofloxacin resistance in *S. Typhimurium* isolates [12,51,54]. Additionally, we reported two new SNVs in MDR efflux pumps related to cephalosporin and fluoroquinolone resistance. Previously, new SNVs in AMR chromosomal genes were reported to be associated with resistance in multidrug and fluoroquinolone [12].

The demand for guinea pigs is growing and becoming widespread, with a high protein and economic value for the Andean countries. Hence, it is one of few genomic surveillance of antimicrobial resistance in isolates of *S. Typhimurium* from guinea pigs in South America. Thus, this study elucidated resistance mechanisms, plasmids' evolution, and the structure of a critical foodborne pathogen. Nonetheless, we still could not define if isolates from meat and farm come from the same contamination source, which would require large and representative data pointing transmission route of *S. Typhimurium*. Furthermore, due to reduced genomic surveillance of foodborne pathogens in Peru, continuous monitoring of the most critical *Salmonella* serovar Infantis, Enteritidis, and Typhimurium from foodborne is needed to monitor outbreaks and potential reservoirs.

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Institutional review board statement

The animal study protocol was approved by the Committee of Animal Ethics and Welfare from the Faculty of Veterinary Medicine of the Universidad Nacional Mayor de San Marcos (protocol code 2016-005, 20-12-2016).

Informed consent statement

Not applicable.

Data sharing

Genome sequence data are available on the National Center for Biotechnology Information (NCBI). Metadata for samples, including accession, were included in Table 1.

Author contributions

R.H., V.A. conceptualized and designed the study; D.E.C.H., L.M.H. collected the isolates data; R.H. methodology design, analyzed the data, interpreted results, and wrote the manuscript; L.L., D.B., M.V.C.V., S.T., F.F.A. edited the manuscript; V.A., L.M.H., B.B.

acquired the funding. V.A. supervised and provided resources and infrastructure for all bioinformatics analyses. All authors read and approved this final manuscript.

Declaration of competing interest

The authors declare no conflict of interest.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.resmic.2023.104086>.

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